

04_event_window_descriptive_diagnostics

May 11, 2026

1 04 Event-Window Descriptive Diagnostics

This notebook runs exploratory monthly event-window diagnostics on the merged salience-polling panel.

Scope: - Descriptive only. - No causal claims. - No event-study regression or significance testing as the main goal. - Objective is to inspect pre/post patterns around key shock months.

```
[1]: from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

pd.set_option('display.max_columns', 200)
pd.set_option('display.width', 180)

PROJECT_ROOT = Path.cwd()
if PROJECT_ROOT.name == 'notebooks':
    PROJECT_ROOT = PROJECT_ROOT.parent

PANEL_DIR = PROJECT_ROOT / 'data' / 'panel'
FIGURES_DIR = PROJECT_ROOT / 'figures' / '04_notebook_figures'
OUTPUT_TABLE_DIR = PANEL_DIR

FIGURES_DIR.mkdir(parents=True, exist_ok=True)
OUTPUT_TABLE_DIR.mkdir(parents=True, exist_ok=True)

PANEL_PARQUET_PATH = PANEL_DIR / 'monthly_salience_polling_panel.parquet'
PANEL_CSV_PATH = PANEL_DIR / 'monthly_salience_polling_panel.csv'
EVENT_SUMMARY_CSV_PATH = OUTPUT_TABLE_DIR / 'event_window_descriptive_summary.
↳csv'

ANALYSIS_START = pd.Timestamp('2017-01-01')
ANALYSIS_END = pd.Timestamp('2025-12-31')
EXPECTED_ROWS = 108

print('PROJECT_ROOT:', PROJECT_ROOT)
```

```

print('PANEL_PARQUET_PATH:', PANEL_PARQUET_PATH)
print('EVENT_SUMMARY_CSV_PATH:', EVENT_SUMMARY_CSV_PATH)
print('FIGURES_DIR:', FIGURES_DIR)

```

```

PROJECT_ROOT: c:\PythonProjects\AfD-project
PANEL_PARQUET_PATH: c:\PythonProjects\AfD-
project\data\panel\monthly_salience_polling_panel.parquet
EVENT_SUMMARY_CSV_PATH: c:\PythonProjects\AfD-
project\data\panel\event_window_descriptive_summary.csv
FIGURES_DIR: c:\PythonProjects\AfD-project\figures\04_notebook_figures

```

```

[2]: def load_panel() -> pd.DataFrame:
    if PANEL_PARQUET_PATH.exists():
        return pd.read_parquet(PANEL_PARQUET_PATH)
    if PANEL_CSV_PATH.exists():
        return pd.read_csv(PANEL_CSV_PATH)
    raise FileNotFoundError(f'Missing panel files: {PANEL_PARQUET_PATH} and
↪{PANEL_CSV_PATH}')

panel = load_panel().copy()
panel['month_dt'] = pd.to_datetime(panel['month_dt'], errors='coerce')
panel['year_month'] = panel['month_dt'].dt.strftime('%Y-%m')

required_cols = [
    'year_month', 'month_dt', 'afd_poll_support_mean', 'afd_poll_support_z',
    'nexis_total_results', 'media_salience_volume_z', 'n_polls'
]
missing_cols = [c for c in required_cols if c not in panel.columns]
if missing_cols:
    raise ValueError(f'Missing required columns: {missing_cols}')

if panel['year_month'].duplicated().any():
    dupes = panel.loc[panel['year_month'].duplicated(), 'year_month'].tolist()
    raise ValueError(f'Duplicate year_month rows: {dupes[:10]}')

if panel['month_dt'].isna().any():
    raise ValueError('month_dt contains unparseable values')

panel = panel.sort_values('month_dt').reset_index(drop=True)

if panel['month_dt'].min().to_period('M') != ANALYSIS_START.to_period('M') or
↪panel['month_dt'].max().to_period('M') != ANALYSIS_END.to_period('M'):
    raise ValueError(
        f'Unexpected panel month coverage: {panel["month_dt"].min()} to
↪{panel["month_dt"].max()} '
        f'(expected months {ANALYSIS_START.to_period("M")} to {ANALYSIS_END.
↪to_period("M")})')

```

```

)

if len(panel) != EXPECTED_ROWS:
    raise ValueError(f'Unexpected panel row count: {len(panel)} (expected_{EXPECTED_ROWS})')

print('panel shape:', panel.shape)
print('coverage:', panel['month_dt'].min(), 'to', panel['month_dt'].max())
print('required columns validated')
print(panel[required_cols].head())
print('Basic summary:')
print(panel[['afd_poll_support_mean', 'nexus_total_results', 'media_salience_volume_z', 'n_polls']].describe())

```

panel shape: (108, 34)

coverage: 2017-01-01 00:00:00 to 2025-12-01 00:00:00

required columns validated

	year_month	month_dt	afd_poll_support_mean	afd_poll_support_z
nexus_total_results			media_salience_volume_z	n_polls
0	2017-01	2017-01-01	12.186667	-0.521216
255			0.494077	15
1	2017-02	2017-02-01	10.356522	-0.909724
239			0.388611	23
2	2017-03	2017-03-01	9.529167	-1.085357
239			0.388611	24
3	2017-04	2017-04-01	9.064286	-1.184043
179			-0.006887	28
4	2017-05	2017-05-01	8.258621	-1.355072
171			-0.059620	29

Basic summary:

	afd_poll_support_mean	nexus_total_results	media_salience_volume_z
n_polls			
count	108.000000	108.000000	108.000000
108.000000			
mean	14.641958	134.787037	-0.298323
23.046296			
std	4.732661	64.178788	0.423043
4.537248			
min	7.896000	44.000000	-0.896758
13.000000			
25%	10.723103	89.250000	-0.598486
20.000000			
50%	13.745238	118.000000	-0.408977
22.000000			
75%	17.511364	168.750000	-0.074452
26.000000			
max	26.022727	353.000000	1.140057
38.000000			

```
[3]: shock_meta = pd.DataFrame([
    {
        'shock_label': 'Chemnitz',
        'shock_month': pd.Timestamp('2018-08-01'),
        'expected_direction': 'ambiguous',
        'mechanism': 'law_order_far_right_mobilization',
    },
    {
        'shock_label': 'Correctiv',
        'shock_month': pd.Timestamp('2024-01-01'),
        'expected_direction': 'decrease',
        'mechanism': 'democratic_threat_remigration',
    },
])

print(shock_meta)
print('Note: Cologne 2016 is outside current DAWUM panel and is not included in
event-window calculations.')
```

	shock_label	shock_month	expected_direction	mechanism
0	Chemnitz	2018-08-01	ambiguous	law_order_far_right_mobilization
1	Correctiv	2024-01-01	decrease	democratic_threat_remigration

Note: Cologne 2016 is outside current DAWUM panel and is not included in event-window calculations.

```
[4]: def extract_event_window(df: pd.DataFrame, shock_month: pd.Timestamp,
    pre_months: int = 6, post_months: int = 6) -> pd.DataFrame:
    start = shock_month - pd.DateOffset(months=pre_months)
    end = shock_month + pd.DateOffset(months=post_months)

    w = df[(df['month_dt'] >= start) & (df['month_dt'] <= end)].copy()
    w['rel_month'] = ((w['month_dt'].dt.year - shock_month.year) * 12 +
    (w['month_dt'].dt.month - shock_month.month)).astype(int)
    w = w.sort_values('month_dt').reset_index(drop=True)
    return w

def compute_pre_post_summary(w: pd.DataFrame, shock_label: str, shock_month: pd.
    Timestamp, pre_months: int, post_months: int) -> dict:
    pre = w[w['rel_month'] < 0]
    post = w[w['rel_month'] > 0]
    shock_row = w[w['rel_month'] == 0]

    afd_shock = float(shock_row['afd_poll_support_mean'].iloc[0]) if
    len(shock_row) == 1 else np.nan
    salience_shock = float(shock_row['media_salience_volume_z'].iloc[0]) if
    len(shock_row) == 1 else np.nan
```

```

out = {
    'shock_label': shock_label,
    'shock_month': shock_month.strftime('%Y-%m'),
    'pre_months': pre_months,
    'post_months': post_months,
    'afd_pre_mean': pre['afd_poll_support_mean'].mean(),
    'afd_post_mean': post['afd_poll_support_mean'].mean(),
    'afd_change_post_minus_pre': post['afd_poll_support_mean'].mean() -
    ↪pre['afd_poll_support_mean'].mean(),
    'salience_pre_mean': pre['media_salience_volume_z'].mean(),
    'salience_post_mean': post['media_salience_volume_z'].mean(),
    'salience_change_post_minus_pre': post['media_salience_volume_z'].
    ↪mean() - pre['media_salience_volume_z'].mean(),
    'afd_shock_month': afd_shock,
    'salience_shock_month': salience_shock,
    'n_months_pre': int(len(pre)),
    'n_months_post': int(len(post)),
}
return out

```

```

[5]: summaries = []
for _, shock in shock_meta.iterrows():
    for win in [3, 6]:
        w = extract_event_window(panel, shock['shock_month'], pre_months=win,
        ↪post_months=win)
        summaries.append(compute_pre_post_summary(w, shock['shock_label'],
        ↪shock['shock_month'], win, win))

event_summary = pd.DataFrame(summaries)
event_summary = event_summary.sort_values(['shock_month', 'pre_months']).
    ↪reset_index(drop=True)

print('Event-window descriptive summary:')
print(event_summary)

event_summary.to_csv(EVENT_SUMMARY_CSV_PATH, index=False)
print('Saved:', EVENT_SUMMARY_CSV_PATH)

```

Event-window descriptive summary:

	shock_label	shock_month	pre_months	post_months	afd_pre_mean	afd_post_mean
0	Chemnitz	2018-08	3	3	15.070079	15.515632
					0.445552	0.421569
					0.199650	-0.221918
1	Chemnitz	2018-08	6	6	14.462367	14.529741
					0.067374	0.109565
					0.135931	0.026367

2	Correctiv	2024-01	3	3	21.508152	18.228030
		0.322694			-0.488077	-0.810771
3	Correctiv	2024-01	6	6	21.172471	17.519438
		0.039254			-0.483682	-0.522936

	afd_shock_month	salience_shock_month	n_months_pre	n_months_post
0	15.463158	0.190862	3	3
1	15.463158	0.190862	6	6
2	21.340000	-0.435344	3	3
3	21.340000	-0.435344	6	6

Saved: c:\PythonProjects\AfD-
project\data\panel\event_window_descriptive_summary.csv

```
[6]: def plot_local_windows(panel_df: pd.DataFrame, shock_label: str, shock_month: pd.Timestamp, window_months: int = 6):
    w = extract_event_window(panel_df, shock_month, pre_months=window_months, post_months=window_months)

    # A) AfD polling around shock
    fig, ax1 = plt.subplots(figsize=(9, 4.8))
    ax1.plot(w['rel_month'], w['afd_poll_support_mean'], marker='o', color='#4f2d2d', linewidth=2, label='AfD polling mean')
    ax1.axvline(0, color='black', linestyle='--', linewidth=1)
    ax1.set_xlabel('Relative month (shock=0)')
    ax1.set_ylabel('AfD polling support (%)')
    ax1.set_title(f'{shock_label}: AfD Polling in Local Event Window ({window_months} months)')
    ax1.grid(alpha=0.25)

    # light n_polls annotation via secondary axis bar
    ax2 = ax1.twinx()
    ax2.bar(w['rel_month'], w['n_polls'], alpha=0.15, color='#1f4e79', width=0.8, label='n_polls')
    ax2.set_ylabel('n_polls (light bars)')

    lines1, labels1 = ax1.get_legend_handles_labels()
    lines2, labels2 = ax2.get_legend_handles_labels()
    ax1.legend(lines1 + lines2, labels1 + labels2, loc='upper left')

    fig.tight_layout()

    name_base = shock_label.lower()
    out_poll = FIGURES_DIR / f'{name_base}_afd_polling_event_window.png'
    fig.savefig(out_poll, dpi=150)
    plt.show()
    print('Saved:', out_poll)
```

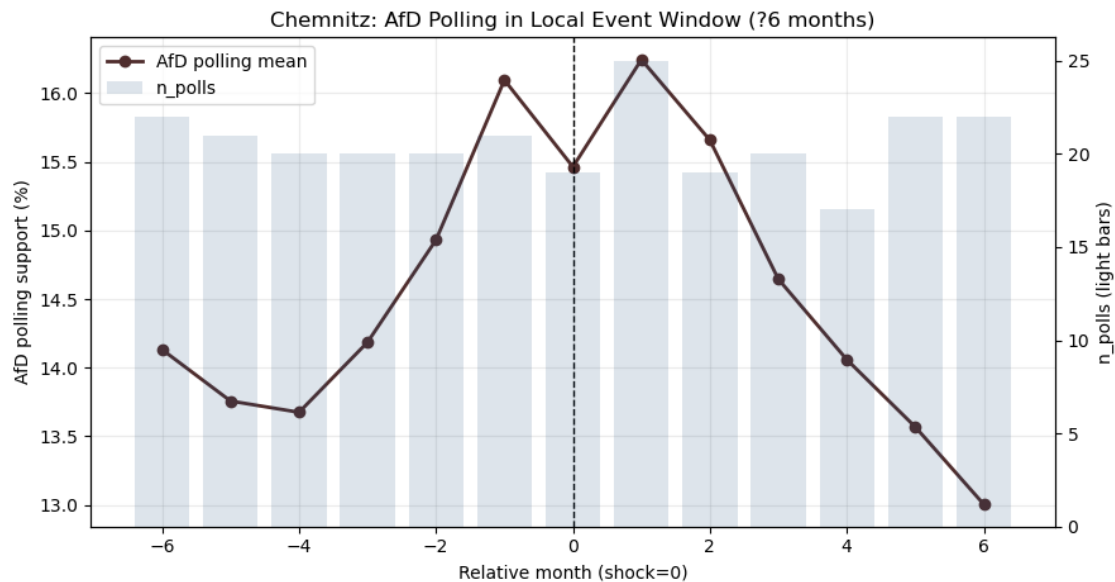
```

# B) Standardized local co-movement
fig, ax = plt.subplots(figsize=(9, 4.8))
ax.plot(w['rel_month'], w['afd_poll_support_z'], marker='o',
color='#8a1c1c', linewidth=2, label='afd_poll_support_z')
ax.plot(w['rel_month'], w['media_salience_volume_z'], marker='o',
color='#1c4b8a', linewidth=2, label='media_salience_volume_z')
ax.axvline(0, color='black', linestyle='--', linewidth=1)
ax.set_xlabel('Relative month (shock=0)')
ax.set_ylabel('Standardized value (z)')
ax.set_title(f'{shock_label}: Standardized Local Co-Movement (?
{window_months} months)')
ax.grid(alpha=0.25)
ax.legend(loc='upper left')

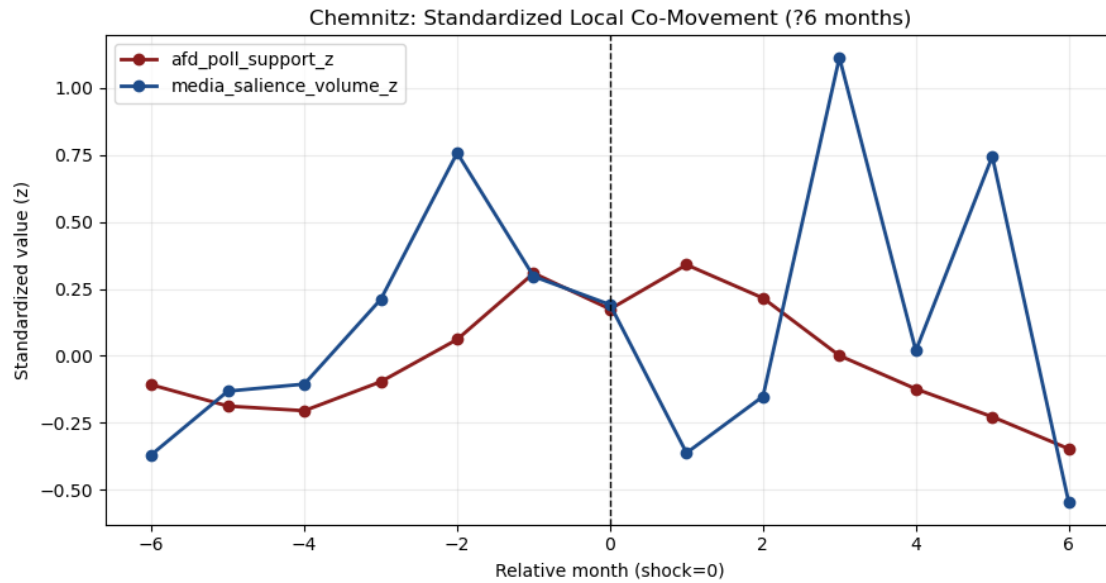
fig.tight_layout()
out_co = FIGURES_DIR / f'{name_base}_comovement_event_window.png'
fig.savefig(out_co, dpi=150)
plt.show()
print('Saved:', out_co)

for _, shock in shock_meta.iterrows():
    plot_local_windows(panel, shock['shock_label'], shock['shock_month'],
window_months=6)

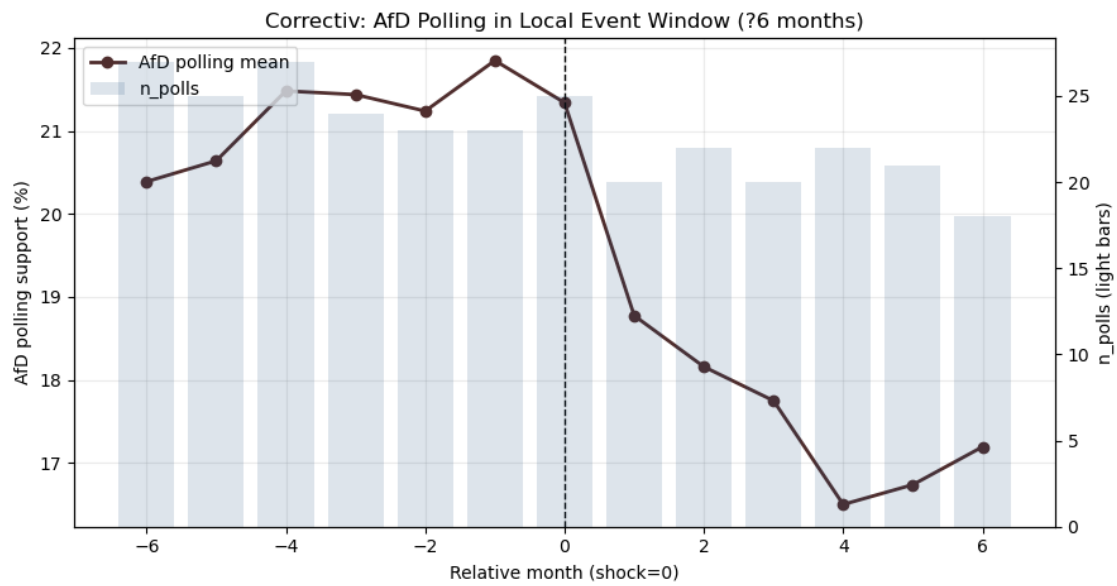
```



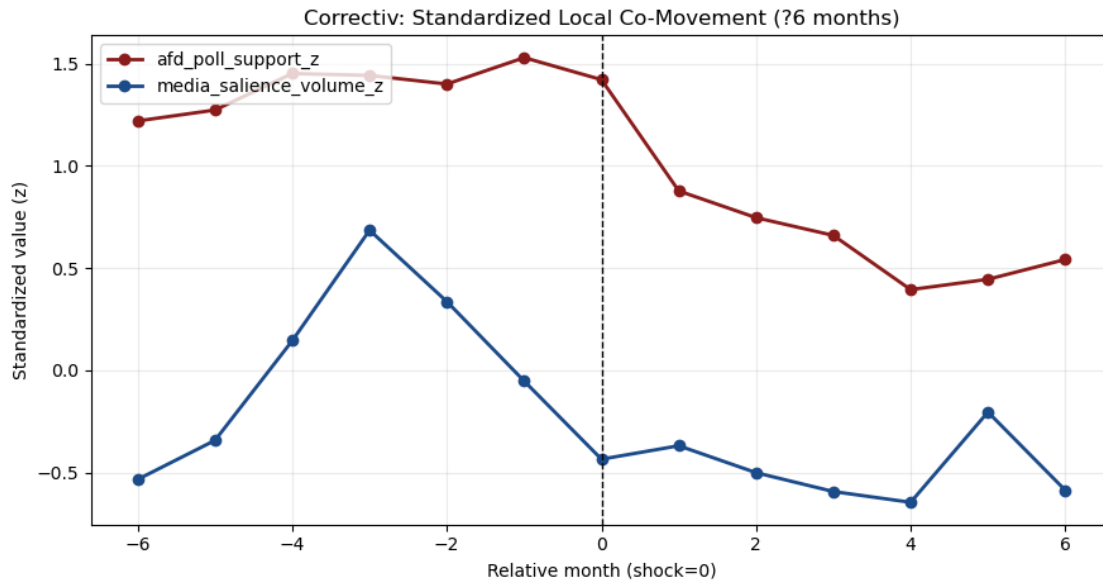
Saved: c:\PythonProjects\AfD-project\figures\04_notebook_figures\chemnitz_afd_polling_event_window.png



Saved: c:\PythonProjects\AfD-project\figures\04_notebook_figures\chemnitz_comovement_event_window.png



Saved: c:\PythonProjects\AfD-project\figures\04_notebook_figures\correctiv_afd_polling_event_window.png



Saved: c:\PythonProjects\AfD-
project\figures\04_notebook_figures\correctiv_comovement_event_window.png

```
[7]: panel_fd = panel.copy()
panel_fd['afd_poll_support_change'] = panel_fd['afd_poll_support_mean'].diff()
panel_fd['media_salience_volume_change'] = panel_fd['nexis_total_results'].
    ↪diff()

fd_cols = [
    'rel_month', 'year_month', 'afd_poll_support_mean',
    ↪'afd_poll_support_change',
    'nexis_total_results', 'media_salience_volume_change',
    ↪'media_salience_volume_z', 'n_polls'
]

for _, shock in shock_meta.iterrows():
    w = extract_event_window(panel_fd, shock['shock_month'], pre_months=6,
    ↪post_months=6)
    print(f"First-difference diagnostic | {shock['shock_label']} (?6 months)")
    print(w[fd_cols].to_string(index=False))
    print()
```

First-difference diagnostic | Chemnitz (?6 months)

	rel_month	year_month	afd_poll_support_mean	afd_poll_support_change	nexis_total_results	media_salience_volume_change	media_salience_volume_z	n_polls
	-6	2018-02	14.131818				1.381818	

124			-102.0	-0.369427		22	
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	-5	2018-03	13.757143		-0.374675
160			36.0	-0.132128	21
	-4	2018-04	13.675000		-0.082143
164			4.0	-0.105762	20
	-3	2018-05	14.185000		0.510000
212			48.0	0.210636	20
	-2	2018-06	14.930000		0.745000
295			83.0	0.757742	20
	-1	2018-07	16.095238		1.165238
225			-70.0	0.296328	21
	0	2018-08	15.463158		-0.632080
209			-16.0	0.190862	19
	1	2018-09	16.244000		0.780842
125			-84.0	-0.362836	25
	2	2018-10	15.657895		-0.586105
157			32.0	-0.151903	19
	3	2018-11	14.645000		-1.012895
349			192.0	1.113690	20
	4	2018-12	14.058824		-0.586176
183			-166.0	0.019479	17
	5	2019-01	13.568182		-0.490642
293			110.0	0.744559	22
	6	2019-02	13.004545		-0.563636
97			-196.0	-0.547401	22

First-difference diagnostic | Correctiv (?6 months)

	rel_month	year_month	afd_poll_support_mean	afd_poll_support_change	nexis_total_results	media_salience_volume_change	media_salience_volume_z	n_polls
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	-6	2023-07	20.388889		1.210317
99			-31.0	-0.534218	27
	-5	2023-08	20.640000		0.251111
128			29.0	-0.343061	25
	-4	2023-09	21.481481		0.841481
202			74.0	0.144720	27
	-3	2023-10	21.437500		-0.043981
284			82.0	0.685234	24
	-2	2023-11	21.239130		-0.198370
231			-53.0	0.335878	23
	-1	2023-12	21.847826		0.608696
172			-59.0	-0.053029	23
	0	2024-01	21.340000		-0.507826
114			-58.0	-0.435344	25
	1	2024-02	18.775000		-2.565000
124			10.0	-0.369427	20
	2	2024-03	18.159091		-0.615909
104			-20.0	-0.501260	22
	3	2024-04	17.750000		-0.409091

90			-14.0		-0.593543	20
	4	2024-05		16.500000		-1.250000
82			-8.0		-0.646276	22
	5	2024-06		16.738095		0.238095
149			67.0		-0.204636	21
	6	2024-07		17.194444		0.456349
91			-58.0		-0.586951	18

1.1 Interpretation Notes (Descriptive)

- Chemnitz is theoretically relevant but descriptively ambiguous in a volume-based salience framework.
- Correctiv is descriptively promising for a potential negative polling-shift pattern if AfD support declines after 2024-01.
- Broad migration/public-order salience may not fully capture mechanism-specific framing; later work may require content/framing indicators.
- Cologne is outside the current DAWUM panel and requires a historical polling supplement for quantitative inclusion.
- These diagnostics motivate next checks but do not establish causal effects.